
Tutorial #3

Topics covered:

- Closed sets and limit points

Problem 1

Let A, B , and A_α denote subsets of a topological space X . Prove the following:

- (a) If $A \subset B$, then $\bar{A} \subset \bar{B}$.
- (b) $\overline{A \cup B} = \bar{A} \cup \bar{B}$.
- (c) $\overline{\bigcup A_\alpha} \supset \bigcup \bar{A}_\alpha$; give an example where equality fails.

Problem 2

Let A, B , and A_α denote subsets of a topological space X . Determine whether the following equations hold; if an equality fails, determine whether one of the inclusions $\supset$ or $\subset$ holds.

- (a) $\overline{A \cap B} = \bar{A} \cap \bar{B}$.
- (b) $\overline{\bigcap A_\alpha} = \bigcap \bar{A}_\alpha$.
- (c) $\overline{A - B} = \bar{A} - \bar{B}$.

Problem 3

Let $A \subset X$ and $B \subset Y$ where X and Y are topological spaces. Show that in the space $X \times Y$,

$$\overline{A \times B} = \bar{A} \times \bar{B}.$$

Problem 4

Show that the product of two Hausdorff topological spaces is Hausdorff.

Problem 5

Show that a subspace of a Hausdorff topological space is Hausdorff.

Problem 6

If $A \subset X$, we define the boundary of A by the equation

$$\text{Bd } A = \bar{A} \cap \overline{(X - A)}.$$

(a) Show that $\text{Int } A$ and $\text{Bd } A$ are disjoint, and $\bar{A} = \text{Int } A \cup \text{Bd } A$.

(b) Show that $\text{Bd } A = \emptyset \Leftrightarrow A$ is both open and closed.

(c) Show that U is open $\Leftrightarrow \text{Bd } U = \bar{U} - U$.

(d) If U is open, is it true that $U = \text{Int}(\bar{U})$? Justify your answer.

Problem 7

Find the boundary and the interior of each of the following subsets of $\mathbb{R}^2$:

(a) $A = \{(x, y) \mid y = 0\}$

(b) $B = \{(x, y) \mid x > 0 \text{ and } y \neq 0\}$

(c) $C = A \cup B$

(d) $D = \{(x, y) \mid x \text{ is rational} \}$

(e) $E = \{(x, y) \mid 0 < x^2 - y^2 \leq 1\}$

(f) $F = \{(x, y) \mid x \neq 0 \text{ and } y \leq 1/x\}$